25. Packages: dnf/rpm (RHEL/Fedora) and Alpine apk

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Phase: Phase 8 — Packages, disks, storage
Estimated time: 1 day
Slug: `25-packages-dnf`

Learning objectives

- Map apt concepts to dnf and apk
- Install/query/remove packages
- Know when distroless/minimal images help
- Read release notes before major upgrades

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics. Type every command yourself — muscle memory beats skimming.

Why this matters

Linux is not one package system. Debian and Ubuntu use `apt`, Red Hat, Fedora, Rocky, and Alma use `dnf`, and Alpine (the base of many small containers) uses `apk`. If you move between laptops, cloud images, and containers, you must translate quickly: “the install command here is X, the package name there is Y”. Tutorials usually show only one family.

This lesson is a translation table plus the extra concepts: repositories, GPG keys, groups, and which package provides a given file.

Core concepts

Task	Debian / Ubuntu	RHEL / Fedora	Alpine
Refresh meta-data	`apt update`	`dnf makecache` (auto)	`apk update`
Install	`apt install X`	`dnf install X`	`apk add X`
Remove	`apt remove X`	`dnf remove X`	`apk del X`
Upgrade all	`apt upgrade`	`dnf upgrade`	`apk upgrade`
Search	`apt search X`	`dnf search X`	`apk search X`
Info	`apt show X`	`dnf info X`	`apk info X`
List installed	`dpkg -l`	`rpm -qa`	`apk info`
File owner	`dpkg -S /path`	`rpm -qf /path`	`apk info -W /path`
Which package has file	`apt-file search`	`dnf provides '*/file'`	`apk info -W /` pkgs site
Local package file	`dpkg -i x.deb`	`rpm -i x.rpm`	`apk add ./x.apk`
Repo config	`/etc/apt/sources.list.d/`	`/etc/yum.repos.d/`	`/etc/apk/repositories`
C library	glibc	glibc	musl
Init	systemd	systemd	OpenRC

Same job, three dialects:

```
"install curl"
    |
    +-- debian/ubuntu --> apt install curl
    |
    +-- rhel/fedora   --> dnf install curl
    |
    +-- alpine        --> apk add curl

Under the hood:  apt -> dpkg (.deb)
                 dnf -> rpm  (.rpm)
                 apk -> .apk archives
```

Detect the family from `/etc/os-release`:

Field	Example
`ID`	`ubuntu`, `rocky`, `alpine`
`ID_LIKE`	`debian`, `rhel fedora`

Worked examples

Detect the distribution:

```
. /etc/os-release
echo "$ID / ${ID_LIKE:-none}"
```

Sample output:

```
rocky / rhel centos fedora
```

Install and query on Fedora/RHEL:

```
sudo dnf install -y tree
dnf info tree | head -4
rpm -qf /usr/bin/tree
dnf provides '*/bin/dig'
```

Sample output:

```
tree-2.1.0-1.el9.x86_64
bind-utils-9.16.23-1.el9.x86_64 : Utilities for querying DNS name servers
```

Install on Alpine, such as in a container:

```
docker run --rm -it alpine:3.20 sh -c 'apk update && apk add curl && curl --
version | head -1'
```

Sample output:

```
fetch https://dl-cdn.alpinelinux.org/alpine/v3.20/main/x86_64/APKINDEX.tar.gz
(1/9) Installing ca-certificates (20240705-r0)
curl 8.9.1 (x86_64-alpine-linux-musl) libcurl/8.9.1
```

A portable installer function:

```
install_pkg() {
  . /etc/os-release
  case "${ID_LIKE:-$ID}" in
    *debian*|ubuntu|debian) sudo apt-get install -y "$@" ;;
    *rhel*|*fedora*)        sudo dnf install -y "$@" ;;
    alpine)                 apk add --no-cache "$@" ;;
    *) echo "unsupported: $ID" >&2; return 1 ;;
  esac
}
```

The mapping from the exercise:

```
map_install_cmd() {
  case "$1" in
    debian) echo apt ;;
    rhel)   echo dnf ;;
    alpine) echo apk ;;
    *)      return 1 ;;
  esac
}
map_install_cmd rhel
```

Sample output:

```
dnf
```

Mini-lab

1. Run `cat /etc/os-release` and note `ID` and `ID_LIKE`.
2. Start containers: `docker run --rm -it ubuntu:24.04 bash`, `fedora:40 bash`, and `alpine:3.20 sh`.
3. In each, install `curl` with the native command and record the output.
4. Use the file-owner query for `/bin/sh` in each environment.
5. Search for `htop` in each package manager.
6. Compare package names: `apache2` versus `httpd`, `dnsutils` versus `bind-utils`, `build-essential` versus `gcc make`.
7. Write the `map_install_cmd` function and test the three families plus an unknown value.
8. In Alpine, note why `bash` is absent by default and `/bin/sh` is busybox.

Decision guide / failure modes

Symptom	Cause	Fix
`apt: command not found` on Fedora	Wrong family	Use `dnf`
`dnf: command not found` on older CentOS 7	Uses `yum`	Use `yum`
Package name not found	Names differ per distro	Search or use `provides`
`bash: not found` on Alpine	Only `ash` installed	`apk add bash`
Binary fails with “not found” on Alpine	glibc binary on musl	Use a native build or Debian image
Repository key error	Missing or expired GPG key	Import vendor key

Think in tasks, not commands: refresh, install, remove, search, file owner. Keep a cheat sheet like the table above, and in scripts branch on `/etc/os-release` instead of guessing.

Common mistakes

- Running untrusted RPM scripts without review
- Assuming package names identical across distros
- Upgrading major releases without backups

Exercises

Work in `exercises/25-packages-dnf/`. Read that folder's `README.md` fully before coding.

```
cd exercises/25-packages-dnf
chmod +x test.sh exercise.sh
./test.sh
```

What you will build

Implement map_install_cmd in exercise.sh. It takes one argument naming a distribution family and prints the name of the native package manager command: debian maps to apt, rhel maps to dnf, and alpine maps to apk. Use a case statement and quote the argument. For an unknown family, print nothing and return 1. Output only the command name, with no other text. Run ./test.sh, then validate the table on real systems or containers by running the actual install command in each family.

Study loop

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** run 2–3 lab commands from this lesson on a real Linux host; paste notable output into your notes
- **C (Challenge):** rewrite the exercise from memory the next day

Do not open `solutions/25-packages-dnf/` until you have an honest failing attempt.

Self-check

Try to answer without scrolling up:

1. dnf equivalent of apt install?
2. Why minimal images?
3. apk is for which distro?

Further reading

- https://dnf.readthedocs.io/